

Structure function analysis of powder beds in additive manufacturing by laser beam melting

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ABSTRACT

Additive manufacturing and specifically laser-based powder bed fusion processes of metals show a rapidly increasing demand in production technology. One key parameter that impacts the quality of the finished part is the condition of the powder bed. It is not surprising that the uniformity of the metal powder distribution has a great influence on the process stability. Caused by the recoating system, the powder may have uneven roughness or unwanted structures such as waviness. One consequence of these conditions may be for example that the fabricated structure contains porosity. In this paper we are proposing an instrument to detect powder bed irregularities in-process, using the so-called structure function. The dimensional data of the powder bed surface can be rapidly evaluated and, moreover, relevant parameters can be obtained with this statistical tool, allowing for in- or off-line process assessment.

1. Introduction

Laser-based powder bed fusion for the additive production of metals, enables the production of components with complex geometry near net shape. In this case, a metal powder is distributed over a substrate plate and selectively melted by a program-controlled laser [1]. By repeating the process in layers (thickness of around 10–100 µm) using a movable construction platform, the desired component is created in almost any shape. There are more than 50 process parameters, adjustable or fixed, such as laser and scanning parameters, powder material and powder bed properties and recoating parameters [2]. Layer uniformity is essential for the way the machine laser interacts with the powder material during the PBF-LB/M [3] process. Due to an erroneous recoating mechanism or sputtering effects, undesirable roughness or waviness of the powder spreading can occur. This may impact the laser absorption, the heat transfer, the melt pool properties and the quality of the finished part.

Many in-process sensing technologies primarily focus on monitoring melt pool signatures. For instance, a CCD (charge-coupled device) camera or a pyrometer for continuous temperature monitoring of the PBF-LB/M are described [4]. Other non-contact methods as for example optical tomography [5] and optical coherence tomography [6] are also applied to PBF-LB/M, to name just a few. A review of process defects and in-situ monitoring methods in metal powder bed fusion is given

with [7]. Powder bed monitoring is proposed to detect irregularities, caused by, for example, a worn recoating system or by unfused powder particles [8–10]. The technology used in these studies applies consumer DSLR (digital single lens reflex) or CCD cameras, which are installed off-axis inside or outside the process chamber. A disadvantage of the method is that only two-dimensional images, similar to simple photographs, are recorded. The sensing of real height distributions of the powder layer thickness is demonstrated in [11] for laser-based powder bed fusion of polymers and in [12] for that of metals. However, the evaluation of the powder deposition based on three-dimensional data requires new intelligent and at the same time fast and reliable methods, in order to precisely assess the powder layer before starting any consolidation.

The idea proposed in this paper is the implementation of a highly sensitive evaluation method for powder bed roughness and waviness based on the so-called structure function (SF). Therefore, the paper aims to present the SF as an useful analysis tool for assessing the uniformity of the powder distribution in order to prevent process deviations. To describe the spatial arrangement of a surface, either the autocovariance function (ACVF), or its normalized form the autocorrelation function (ACF), or the power spectral density function (PSDF) can be used. However, the application of the structure function to surface height data has proven to provide several advantages over other statistical methods [13]. The SF can be applied to any aperture,

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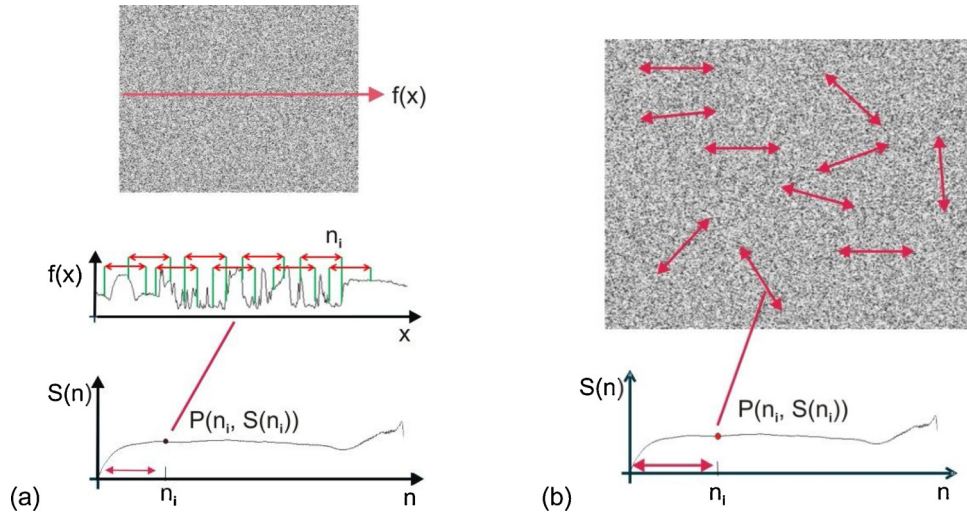


Fig. 1. (a) Construction of the one-dimensional (linear) SF for a profile cut $f(x)$. According to Eq. (3), the calculation for a separation n_i gives one point P of the SF. (b) One-dimensional (linear) SF created multidirectional for the entire area. The statistical expectation of all possible point pairs for a separation n_i here yields one value P of the SF too.

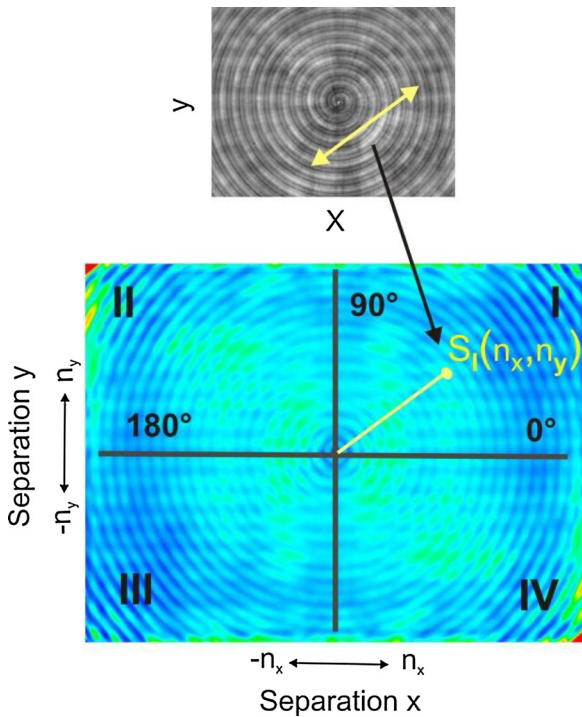


Fig. 2. Construction of the area structure function. In the upper part of the image, the input (object) data are shown. Below, the calculated ASF in all 4 quadrants is visualized. By vector-like scanning (pairs of points have defined distances and specific directions: yellow marked line with arrow tips) of the object data, the ASF for the quadrant I is formed in the example (scanning direction 0° - 90°). The averaging over all possible data pairs yields the structure function $S_i(n_x, n_y)$ according to Eq. (5), as shown in the lower part (quadrant I) of the picture. Similarly, considering Eq. (6), the ASF will be formed for the quadrant II. The squaring of the data in the formation of the ASF results in $S(n_x, n_y) = S(-n_x, -n_y)$ as well as $S(-n_x, n_y) = S(n_x, -n_y)$. Accordingly, the projection follows from quadrant I to III and from quadrant II to IV.

area size or geometry. Data do not have to be pre-processed for evaluation purposes. It is much more sensitive to differences in roughness than ACF or PSDF. The SF that represents the mean height difference of each spatial distance (separation) does not refer to the statistical expectation of the data and is therefore independent of the measurement

tool.

The structure function was originally introduced by Kolmogorov in 1941 to analyze statistical problems associated with the turbulence theory [14,15]. In the meantime, it has been applied in diverse areas: Among others, it has been used for the characterization of technical surfaces [16,17], optical surfaces [18,19], velocity fields in turbulent flows [20], and phase and frequency instabilities in frequency sources [21]. Moreover, it has been used for time series analysis in astronomy [22]. A first approach to inspect powder bed surfaces is given in [23]. In the following, the procedure is presented in detail and the evaluation by means of the structure function is explained comprehensively.

2. Definition of the structure function

The starting point is the calculation of the autocovariance function (ACVF) for a zero-mean discrete data set x_n of finite length N [24]

$$R_{xx}(k) = \frac{1}{N} \sum_{n=0}^{N-1-k} x_{n+k} x_n \quad (1)$$

The index k is called the separation or lag of x_n . It applies to non-negative lags $\{k = 0, \dots, N-1\}$. The normalization of the ACVF with the variance σ_x^2 yields the autocorrelation function (ACF)

$$C_{xx}(k) = \frac{R_{xx}(k)}{\sqrt{\sigma_x^2 \sigma_x^2}} = \frac{R_{xx}(k)}{\sigma_x^2} \quad (2)$$

The structure function of order 2, depending on the separation parameter n for a one-dimensional discrete set of height values z_i , $\{i = 1, \dots, N\}$, is defined by

$$S(n) = \frac{1}{N-n} \sum_{i=1}^{N-n} (z_i - z_{i+n})^2 \quad (3)$$

We can identify the relation between ACF and SF (for non-negative $k = n$) [25] with

$$S(n) = 2\sigma^2[1 - C_{xx}(n)] \quad (4)$$

Due to this close relation, there are similarities between ACF and SF. For example, the ACF of a periodic signal is also periodic. The same behavior applies to the structure function [26]. Despite the correlation according to Eq. (4), the SF has several advantages over the ACF. Since, in principle, only differences in height values are considered, the SF does not need any signal without a mean value. Accordingly, data from different measurement methods can be compared. One of the most

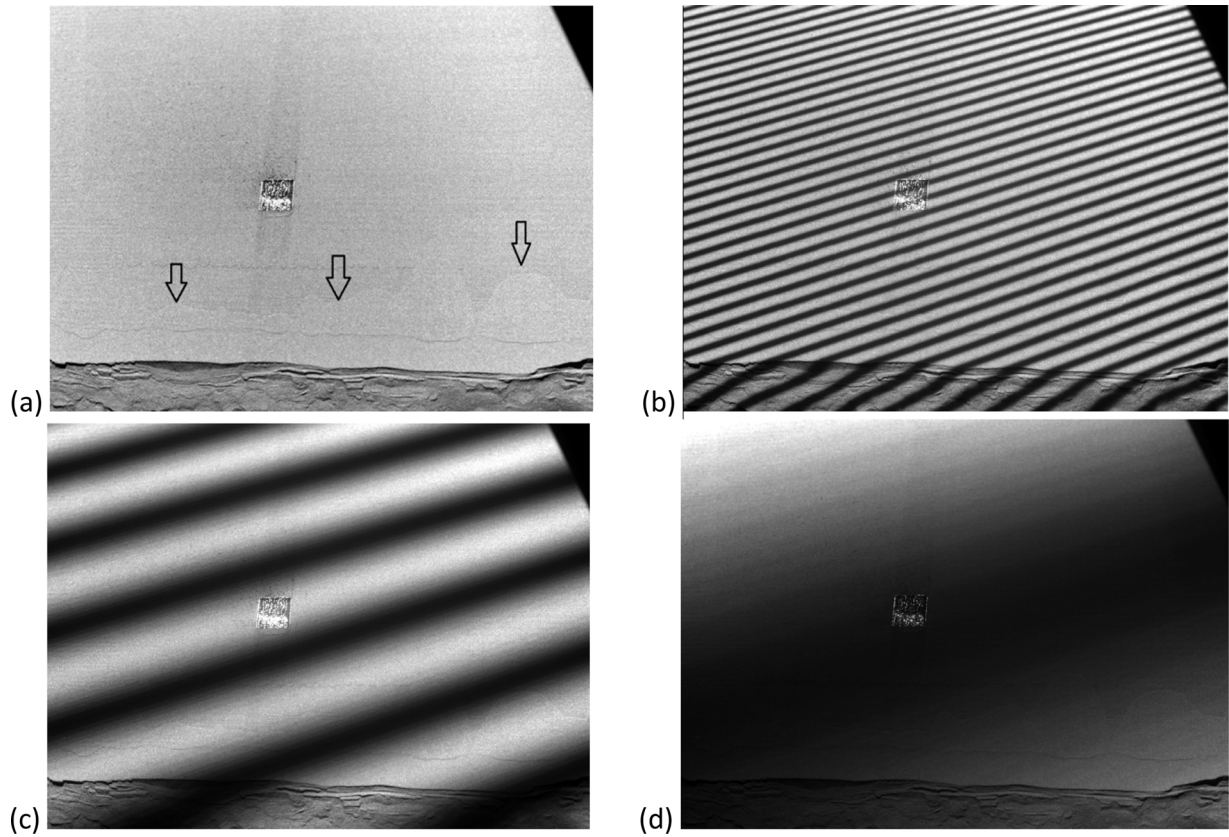


Fig. 3. (a) Camera image taken after a consolidation procedure. The recoating takes place from top to bottom. At first sight, the powder distribution looks uniform. Only a slight layering (marked with black arrows) at the end of the powder bed is visible. At the bottom of the picture it can be noticed that the powder bed piles up, caused by the end of the build plate. The dimension of the measurement area (x, y) is $15 \times 15 \text{ cm}^2$. This corresponds to the field of view of the camera in the picture. (b)-(d) A set of sinusoidal fringe patterns projected onto a powder bed area. The images represent a part of the measurement sequence (for more detailed information see [12]).

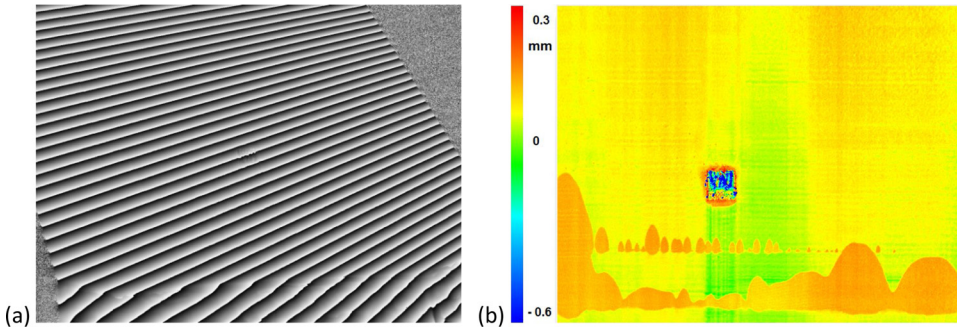


Fig. 4. (a) Wrapped phase value image calculated from fringe pattern shown in Fig. 3(b) to (d). (b) Calculated height value distribution of the powder bed, transformed perpendicular to the (x, y) plane ($15 \times 15 \text{ cm}^2$). A perfect powder spreading should result in a color representation in green only. The irregularities caused by improper recoating procedure are explained in Fig. 5.

important advantages is the significantly higher sensitivity of the SF for roughness differences, which is demonstrated in this paper by applying the SF to powder bed evaluation. Fig. 1 illustrates the construction of the SF according to Eq. (3).

According to Fig. 1(a), each separation n_i of the data set gives a value $S(n)$. However, all possible separations do only for this one direction of the profile section result in the complete SF. This construction can be done for x , y or for any other orientation, retaining the anisotropy of the surface structure. The method can be applied in parallel to the entire surface. The multidirectional design captures the statistical expectation of the SF for each separation by sampling all data in all possible orientations, see Fig. 1(b). This averaging produces a SF with a particularly low noise and simultaneous loss of anisotropy information. In the case of stochastic surfaces, both methods must lead to the same result. It depends on the application which technique is ultimately more beneficial.

Fig. 2 shows another possibility to statistically evaluate surfaces by means of a structure function for a two-dimensional case. One can say, this approach creates a kind of "vector-like" scan of the entire surface. All height differences of a separation of the direction (n_x, n_y) are averaged. All possible orientations of all separations yield the so-called area (or surface) structure function (ASF)

$$S_f(n_x, n_y) = \frac{1}{(M-n_x)(N-n_y)} \sum_{i=1}^{M-n_x} \sum_{j=1}^{N-n_y} (z_{i,j} - z_{i+n_x, j+n_y})^2 \quad (5)$$

The formulation according to Eq. (5), as also found in [27], applies strictly and only to the mathematically first quadrant in a Cartesian coordinate system. A more complete description additionally requires the construction of the 2nd quadrant (separations $(-n_x, n_y)$) for the correct representation with anisotropy of the input data. According to [28] the so-called 2nd quadrant area structure function follows

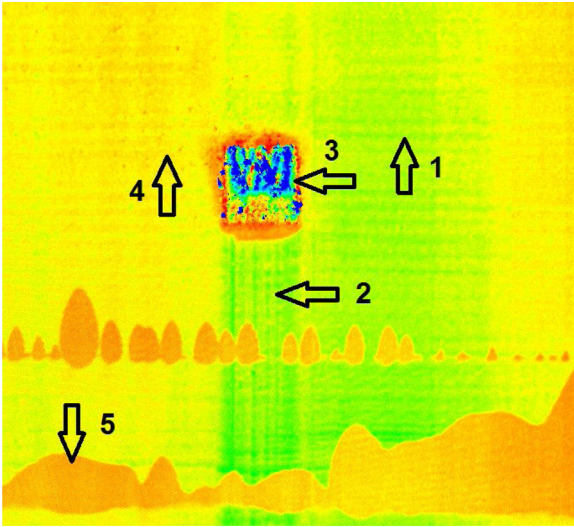


Fig. 5. Cut out (around $10 \times 10 \text{ cm}^2$) of Fig. 4(b) with denoted areas marked with arrows (same color scale as in Fig. 4(b)). (1) Waviness in y-direction, (2) Waviness in x-direction, (3) Non-uniform top layer of consolidation, (4) Spattering area, (5) Formation of piled powder.

$$S_{II}(-n_x, n_y) = \frac{1}{(M-n_x)(N-n_y)} \sum_{i=1+n_x}^M \sum_{j=1}^{N-n_y} (z_{i,j} - z_{i-n_x,j+n_y})^2 \quad (6)$$

The following properties show the importance of the structure function for surface characterization:

- 1 The structure function applied to a surface represents the expectation of the quadratic difference of height fluctuations in time or space as a function of separation (spatial difference).
- 2 The structure function can be interpreted as a criterion of the frequency content of the surface with respect to roughness and waviness.
- 3 Periodicities within the sample surface are again represented by periodicities in the structure function.
- 4 The structure function is independent of the mean, since height differences are calculated.
- 5 Roughness and waviness of fundamentally different data sets can be compared.

3. Powder bed evaluation with structure function analysis

The performance of the structure function for evaluating height data with respect to roughness and waviness, is demonstrated by way of example. As already mentioned above, the uniform distribution of the metal powder on the building plate is an essential criterion for the quality of the laser beam melting process. As with a solid, one can detect and investigate a kind of roughness and waviness in the powder distribution. In this example we use a dataset obtained with a high-precision measuring method based on the fringe projection method. In an extended research work we demonstrated the successful application of dimensional monitoring in-line for the PBF-LB/M process. For further explanation of measurement method and implementation, we refer to [12].

The sample under test (spread metallic alloy powder and/or laser consolidated parts) will be coded with structured light. The powder material used in this work is nickel-chromium based alloy Inconel® 718. The surface profile is obtained from the evaluation of the deformed fringe pattern following the object shape. These fringe patterns are recorded using two CCD cameras. Fig. 3(a) gives an impression of the sample used here. This texture image can be compared to a photographic image. It shows the camera view during measurement. A consolidated cube with a size of approximately $10 \times 10 \text{ mm}^2$ is visible inside the powder bed (size $15 \times 15 \text{ cm}^2$). The data acquisition takes place immediately after the completion of a layer. Thus, a one-dimensional sinusoidal fringe pattern distribution is projected onto the surface with three different wavelengths as part of the measurement sequence [29] (Fig. 3(b)-(d)). The measurement result, a modulated phase image, is calculated with one of the numerous techniques for analyzing fringe patterns [30].

Fig. 4(a) shows the calculated phase values that contain the information of the surface contour. With a calibration of the set-up, the unwrapped phase distribution can be transformed to (x, y, z) coordinates. The accuracy of the system is qualified to be below $10 \mu\text{m}$ (z-coordinate). The lateral resolution is about $60 \mu\text{m}$ (measurement area (x, y) $15 \times 15 \text{ cm}^2$). Fig. 4(b) illustrates the 3D representation of the measurement result. The coordinates (z) are calculated perpendicular to the (x, y) plane. The powder recoating procedure shown in the picture took place from top to bottom. On a first glance, one can notice a waviness of the powder from left to right. The structure at the bottom of the sample results from a deliberately induced uneven powder distribution: There is practically a pile of powder at the end of the recoating process. Furthermore, one can notice a trace in the powder in the width of the consolidation. The colors represent the height values (the scale is given in millimeter).

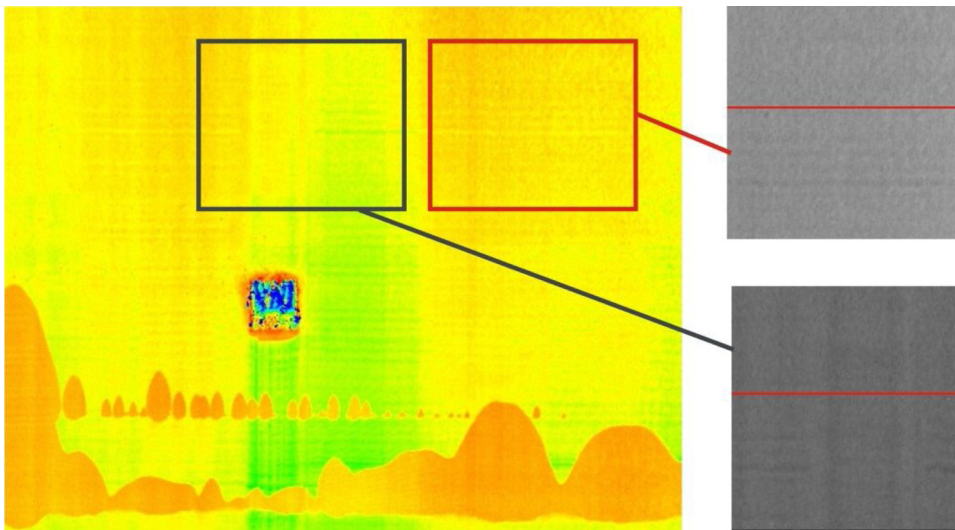


Fig. 6. The measured powder bed has a size of about $15 \times 15 \text{ cm}^2$ (x, y). For area 1 (black frame): PV (Peak to Valley) = $140 \mu\text{m}$, $1 \sigma = 13 \mu\text{m}$, suggests slight ripples in the gray values. For area 2 (red frame): PV = $175 \mu\text{m}$, $1 \sigma = 16 \mu\text{m}$, but appears rather smooth. For each area approximately 100,000 height values are obtained and the lateral dimension is approximately $5 \times 5 \text{ cm}^2$.

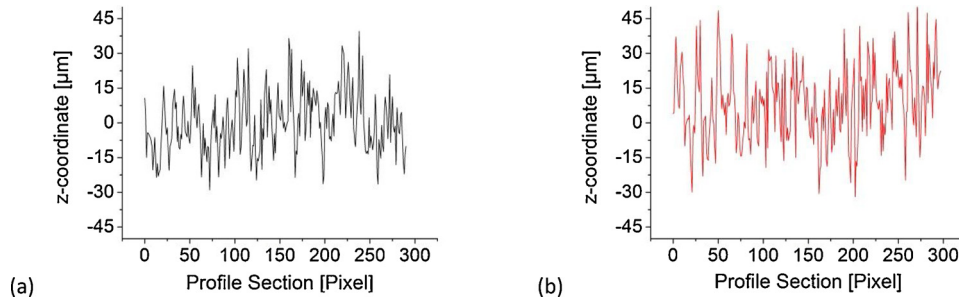


Fig. 7. (a) Line cut within area 1: $R_a = 10 \mu\text{m}$, $R_z = 59 \mu\text{m}$. (b) Line cut within area 2: $R_a = 12 \mu\text{m}$, $R_z = 70 \mu\text{m}$. Both profile sections were taken from the same y-coordinate of the powder bed.

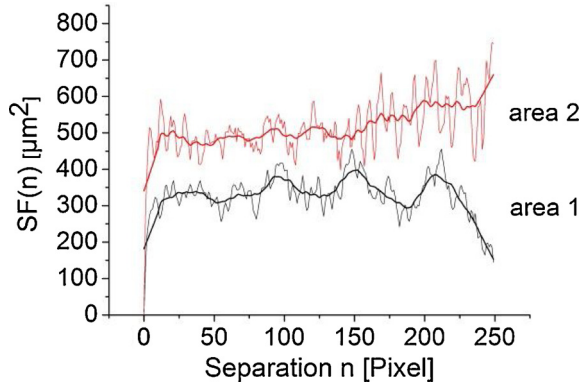


Fig. 8. Calculated linear structure functions of the profile sections from Fig. 7. The SF of the profile measurement of area 1 is drawn in black, the one of area 2 in red. Solid lines show the result of an application of a Savitzky-Golay filtering for both SFs. The difference in roughness and waviness comparing both structure functions is clearly visible.

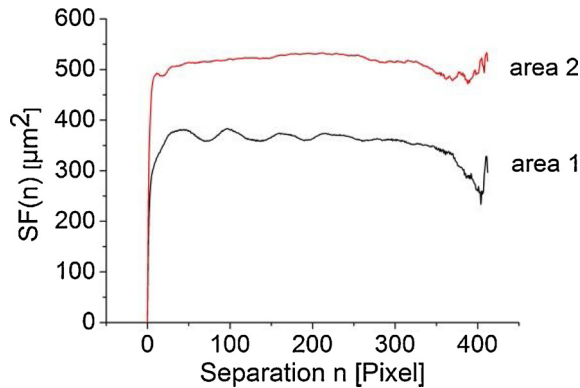


Fig. 9. Multidirectional determined structure functions of area 1 (black) and area 2 (red).

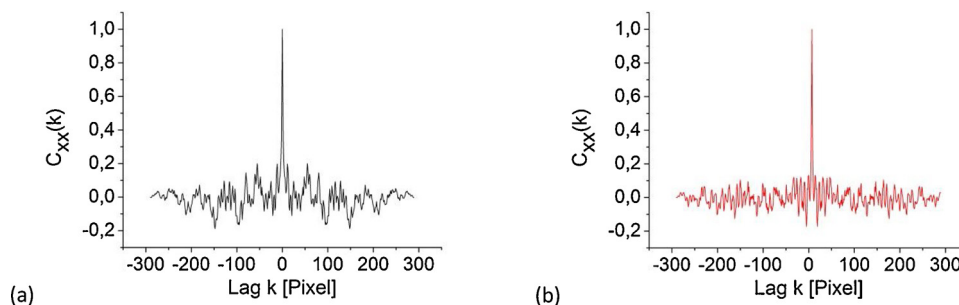


Fig. 10. (a) ACF from the data set of profile section area 1, (b) ACF from the data set of profile section area 2.

In Fig. 5 an enlarged part of the surface depicted in Fig. 4, containing the detected points of failure, is shown. A consequence of the faulty powder distribution is an unequal consolidation. In addition, there is a wave formation in the powder distribution in both the x- and y-direction. Even the spots caused by the spattering effect are visible.

Fig. 6 shows two sections of the measured data of the powder bed in more detail. The distribution in the black frame is named area 1 in the following, the one in the red frame is named area 2. A first quality criterion may be the calculation of the standard deviation of all height values of a surface. Area 1 ($1\sigma = 13 \mu\text{m}$) and area 2 ($1\sigma = 16 \mu\text{m}$) differ only slightly in this respect. It can now be shown that when using the structure function, both surfaces can be assessed and compared in terms of roughness and waviness.

Fig. 7 shows the data of the profile sections of both areas 1 and 2, which are indicated with red lines in the areas shown in Fig. 6. To obtain a simpler assignment of the data in the following results to the above-mentioned areas, the graphs are also drawn in different colors. The raw data of the measured altitude values still do not allow any assessment regarding the powder distribution. The specified geometric characteristics (arithmetic mean roughness R_a , average roughness depth R_z) of the surface finish were determined in accordance with ISO 4287 [31] and ISO 4288 [32]. The results for the profile sections are $R_a = 10 \mu\text{m}$, $R_z = 59 \mu\text{m}$ for area 1, and $R_a = 12 \mu\text{m}$, $R_z = 70 \mu\text{m}$ for area 2.

For both data sets of the profile sections from Fig. 7, the linear structure functions were calculated (compare Fig. 1(a)) and the result is given in Fig. 8. The data of area 2 show higher values than those of area 1 for all separations. Accordingly, the roughness of the powder distribution must be lower for area 1 than it is for area 2. This can be explained with R_a and R_z . Savitzky-Golay filters [33] are performed for both SFs and the results are depicted in the graphs. Area 2 shows a slight ripple, which appears more stochastic (red curve). The data of the profile section of area 1, however, show a distinct waveform across all separations (black curve).

In contrast to Fig. 8, Fig. 9 shows a remarkable difference in the results after applying the linear structure function to all data of both areas. The multidirectional evaluation of all possible point pairs for a separation results in the averaging of the SFs across the areas (compare

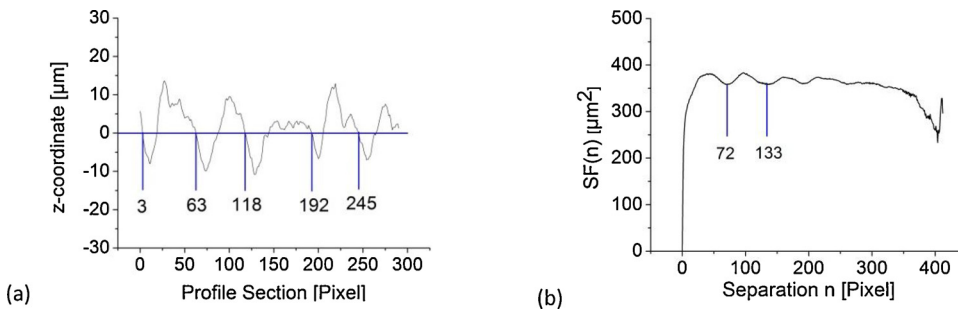


Fig. 11. (a) Lowpass filtered profile data of Fig. 7(a). The distances between the zero-crossings are evaluated according to DIN 4287 and amount to 60.5 pixels on average (spacing parameter WSm). (b) Linear structure function of area 1, see Fig. 9, with marked valley points of the first two minima (distance 61 pixels). Compared to the evaluation according to (a), this observation confirms that the spacing parameter WSm can be determined with the SF. Thus, the structure function is an appropriate indicator for waviness.

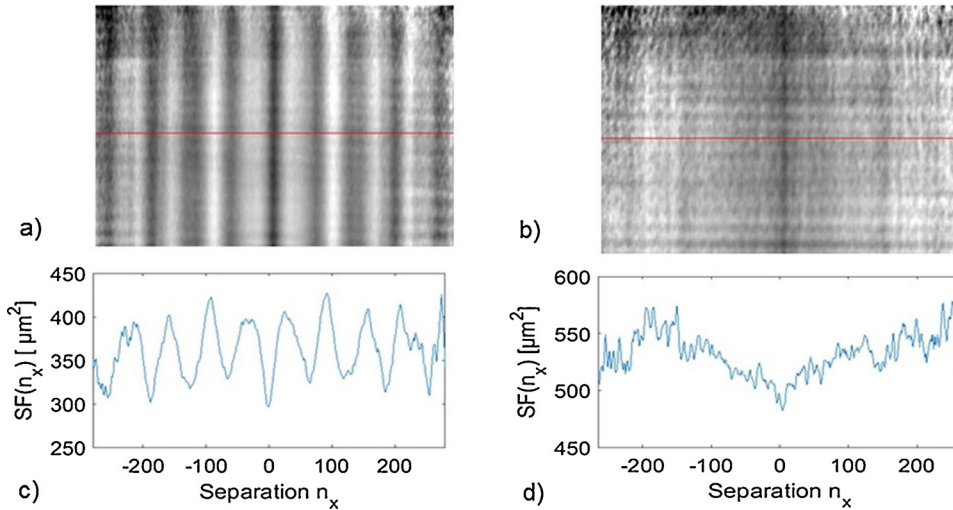


Fig. 12. (a) Calculated area structure function of surface 1. The waveform is clearly visible (same separation range as in Fig. 12(c)). Fig. 12(b) Calculated surface structure function of surface 2. Hardly any ripples present (same separation range as in Fig. 12(d)). Fig. 12(c) line cut of (a) and Fig. 12(d) line cut of (b). The values and forms of the structure functions confirm the results of the linearly calculated SFs (Figs. 8 and 9).

Fig. 1(b)). The structure functions calculated in this way are similar to those in Fig. 8 but very low in terms of noise, and can therefore be assessed relatively simple without further filtering or transformation. It can be clearly recognized that area 1 has a certain periodicity, while the ripple on area 2 is more random and weaker. This is indicated by the rather smooth profile of the obtained SF. The behavior of the SF within each individual profile section is thus also reflected in the entire area. As already mentioned above, the difference between the two methods is that the information about the anisotropy of the surface is retained for profiles (application in one direction only) and is lost for surfaces (multidirectional application).

For comparison, Fig. 10 shows the calculated autocorrelation functions of the same profiles as in Fig. 7. Again, the color assignment corresponds to the graph of both data sets. The calculated ACFs do not allow conclusions concerning differences in roughness in the data sets. The ripple, however, at least in the data set of area 1, can be determined after filtering the signal and further preparation. The presentation of this evaluation is omitted below.

To determine the spacing parameter WSm ("average spacing" designated in accordance with ISO 4287) from the ACF, the maxima of the ACF have to be evaluated. These precisely correspond to the minima of the SF. Fig. 11 shows the result of the determination of the spacing parameter WSm of the SF. Fig. 11(a) shows the evaluation of the low pass filtered profile section of the data from Fig. 7(a) (area 1) according to ISO 4287. The values for the zero crossings of the profile section within the measuring section are shown. The mean groove width in this example is $WSm = 60.5$ pixels. The distance between the two first minima of the SF is 61 pixels (Fig. 11(b)). This confirms that the waviness of the powder bed can be evaluated very precisely by applying the structure function without signal post-processing (e.g. filtering). This means that information about the characteristic waviness of an area under investigation, see e.g. Fig. 11(b), can already be obtained by the evaluation of the first two minima of the SF.

Fig. 12 shows the calculated area structure functions for both area 1 and area 2 and gives evidence of the statements made so far with linear SFs concerning roughness and waviness. The ASFs are determined with Eq. (5) and Eq. (6) and both quadrants are shown side by side. Fig. 12(a) gives the result for area 1. The undulate structure of the surface is clearly visible. Fig. 12(b) shows the result for area 2. As already determined by the linear SF, a wave profile is missing here. The results confirm the previous statements. It should be noted that the "compressed" information of the multidirectional SF usually overrides the anisotropy of the surface data, whereas the ASF does not. In Fig. 12, the information of the SF can be read out and evaluated depending on the line, column or direction. As with the linear SF, the waviness in the surface data can be determined with the ASF. In Figs. 12(c) and 12(d), line sections are shown by way of example.

4. Discussion

We have demonstrated that the structure function provides a relatively easy-to-handle tool for evaluating roughness and waviness. The calculation of the area structure function with a commercially available PC takes less than one second for the surfaces shown in the example with approximately 100,000 height values per area.

For evaluating the powder distribution, one can define an appropriate threshold value for the structure function, as for example a value of less than $400 \mu m^2$ for the desired quality of recoating smoothness in the presented work. Thus, the distribution of area 1, even with the existing ripple, would be within the acceptable range for the process under discussion. Area 2, however, would not be. If waviness occurs, it will also be possible to evaluate minima or maxima.

Our investigations thus indicate that structures with roughness or waviness, which are overlooked by surface evaluation methods such as autocorrelation or Fourier-based methods, can be detected by applying a structure function-based approach. Our results therefore indicate that

the use of structure functions allows for a reliable and significant analysis of powder bed recoating in PBF-LB/M processes. It will be the subject of further investigation to determine the quantitative deviations in roughness and waviness, which have an impact on the process, from the ideal, even powder distribution.

5. Conclusion

We propose a new method for the in-process evaluation of additive manufactured parts using PBF-LB/M based on the structure function analysis. We have shown that when comparing metallic powder height distributions, which differ only by 3 μm in their standard deviations, differences in roughness and waviness can be reliably assessed using the structure function. This statistical tool is, compared to other means such as autocorrelation or Fourier-based methods, extremely sensitive to surface roughness. With appropriate computational power, a layer by layer evaluation appears feasible. PBF-LB/M manufacturing could therefore be re-adjusted during processing so that deviations can be avoided. This would, in conclusion, minimize the reject rate and thereby provide a substantial advantage compared to current online monitoring systems.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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